

On the Capacity of Any Communication Network

A. Chawla

Abstract

The capacity of large-scale, heterogeneous, dynamic networks such as the global Internet remains an open problem in network information theory. Traditional transmission-based formulations require enumerating routes, coding strategies, and link-level constraints, leading to conceptual and combinatorial intractability in the general case. In this work, we present a storage-view framework that reframes the problem as the persistence of a high-dimensional codeword representing the complete network state. This formulation is formally equivalent to the transmission view via the storage–communication duality, yet is often far more tractable to model and analyze. We propose a definition of state persistence capacity for arbitrary networks, demonstrate its mathematical equivalence to network capacity, and outline how it can be estimated for the global Internet using empirical measurements and dynamical modeling. This approach opens a pathway toward progress in a domain long considered resistant to general capacity theory.

I. INTRODUCTION

The problem of determining the capacity of the global Internet, considered as an arbitrary, time-varying, heterogeneous network, is emblematic of the limitations of current network information theory. While classical results provide capacity characterizations for special cases such as point-to-point channels, multiple access channels, and certain relay or broadcast networks, no general closed-form or algorithmic solution exists for networks with arbitrary topology, mobility, and adaptive behavior. The monograph of Kim and El Gamal underscores this limitation, offering a comprehensive treatment of known cases while refraining from any claim that the general network capacity problem is solved.

This lack of general theory has fostered a persistent pessimism in the field. The transmission-based view, in which capacity is defined as the supremum of achievable communication rates

between specified sources and destinations under physical and coding constraints, becomes combinatorially unwieldy when the network comprises billions of nodes, dynamic routing, and heterogeneous protocols. The transmission formulation forces explicit enumeration of flows, paths, and link interactions, which becomes intractable as heterogeneity and dynamics increase.

II. STORAGE–COMMUNICATION EQUIVALENCE

It is well known that the capacity of a channel from one spatial point to another can be framed as the capacity of a storage medium from one temporal point to another. In Shannon’s classic formulation, storing information for a duration in a noisy memory or sending it across a noisy wire are mathematically equivalent when the appropriate mapping between spatial and temporal indices is made. The act of transmitting a symbol from node A at time $t = 0$ to node B at time $t = T$ is isomorphic to storing that symbol at a single location for duration T in the presence of noise. In both cases, the operational definition of capacity is the maximum mutual information between the input and the output under allowable input distributions.

Traditionally, this duality has been applied to single links or small subsystems. We extend this equivalence to the global network state as a whole. The following figure illustrates the core idea: a spatial transmission from a source region to a destination region is mapped to a temporal storage problem in which a single high-dimensional state is preserved across time. The mapping is conceptual and operational; it preserves the role of noise, resource constraints, and the optimization over admissible input distributions.

III. STATE PERSISTENCE CAPACITY

Let $X(0)$ denote the complete state of the Internet at $t = 0$, encompassing traffic flows, routing tables, buffer occupancies, protocol states, and link statuses, represented as a high-dimensional vector. The set of all possible network states is the codebook from which a single realization is drawn. Under the deterministic and stochastic mechanisms that govern packet forwarding, congestion control, routing updates, and physical-layer effects, the state evolves to $X(T)$ at a later time T .

Define the state persistence capacity over interval T as

$$C(T) = \sup_{P_{X(0)} \in \mathcal{P}} I(X(0); X(T)), \quad (1)$$

where $\mathcal{P}$ denotes admissible priors on $X(0)$ reflecting resource and operational constraints (for example, bounds on average queue occupancies, protocol state complexities, or energy budgets).

The quantity $C(T)$ captures the maximum information about the initial global state that can be recovered after time T by any observation or decoding procedure that has access to the network at time T . When causal adaptation of local dynamics is permitted, the supremum may be taken over both admissible priors and admissible adaptive strategies that modify the evolution operator in real time subject to constraints.

The intensive or rate-like form of persistence capacity can be obtained by normalizing by time,

$$c = \limsup_{T \rightarrow \infty} \frac{C(T)}{T}, \quad (2)$$

when the limit exists. This intensive capacity is analogous to long-run communication rates but defined with respect to preservation of global state across time rather than explicit flow rates between predetermined source-destination pairs.

IV. MODELING AND ESTIMATION

A flexible modeling framework for network state evolution is a stochastic dynamical system

$$\frac{dX}{dt} = F(X, t) + \eta(t), \quad (3)$$

where F encodes deterministic protocol-driven evolution such as routing table convergence, traffic engineering actions, and control-loop responses, while $\eta(t)$ summarizes stochastic influences including random delays, link failures, attack traffic, and exogenous demand variability. In the passive regime, F is fixed and known up to parameter uncertainty. In the adaptive regime, F may itself be a functional of past states and observations, representing online optimization, control, or machine learned adaptations.

Estimating $C(T)$ requires approximating the joint distribution of $(X(0), X(T))$. Direct estimation is intractable in the raw high-dimensional representation; practical approaches combine principled dimensionality reduction with variational mutual information estimators. Candidate reduction strategies include linear subspace projection, learned embeddings that preserve predictive information, and physically informed coarse-graining. With a surrogate low-dimensional representation $Z = \Phi(X)$ one estimates $I(Z(0); Z(T))$ using techniques such as variational bounds, MINE-style estimators, or Gaussian approximations in the linear regime.

Analytical bounding is possible in tractable families. Consider linear time-varying dynamics with Gaussian priors and noise,

$$X_{t+1} = W(t)X_t + \xi_t, \quad \xi_t \sim \mathcal{N}(0, \Sigma_\xi). \quad (4)$$

Unrolling yields

$$X_T = \Phi_T X_0 + \sum_{k=0}^{T-1} \Phi_{T-1-k} \xi_k, \quad \Phi_T = \prod_{s=0}^{T-1} W(s). \quad (5)$$

For Gaussian $X_0 \sim \mathcal{N}(0, \Sigma_{X_0})$, mutual information admits a closed-form expression,

$$I(X_0; X_T) = \frac{1}{2} \log \frac{\det(\Phi_T \Sigma_{X_0} \Phi_T^\top + \Sigma_{\text{noise}})}{\det(\Sigma_{\text{noise}})}, \quad (6)$$

where Σ_{noise} aggregates the process noise contributions. Optimization over Σ_{X_0} under trace or power constraints yields explicit capacity bounds that reveal dependencies on network mixing (encoded in Φ_T), mobility or topology changes (captured by time variation in $W(t)$), and noise strength.

Beyond linear-Gaussian models, directed information and causal mutual information provide principled measures when adaptation and feedback are present. The optimization over causal strategies can be framed as a constrained control or stochastic optimization problem where the objective is to maximize retained information about $X(0)$ at time T under energy, observability, and complexity constraints on admissible strategies.

V. ADVANTAGES AND LIMITATIONS

The storage view shifts emphasis from combinatorial enumeration of paths and coding strategies to quantifying information persistence of a single global codeword. This reframing reduces dependence on fine-grained topological details and allows the use of mature tools from stochastic dynamical systems, control theory, and high-dimensional statistics. It gives a single, universal scalar (or curve as a function of T) that is comparable across networks and regimes and that directly captures predictability and resilience properties.

Limitations include the high dimensionality of the network state and the need for careful specification of admissible priors and strategy constraints to ensure operational relevance. The storage formulation does not automatically provide constructive transmission schemes in the traditional sense; rather it provides fundamental limits and comparative diagnostics. Linking capacity bounds in the storage view to implementable coding and routing strategies remains an important direction for future work.

VI. IMPLICATIONS

Recasting the capacity of the global Internet as a state persistence problem provides a tractable conceptual foothold in a domain where the transmission formulation has proven resistant to

general solutions. The approach enables empirical estimation from measurement infrastructures, analytical bounding in tractable families, and design of adaptive strategies targeted at maximizing preservation of critical global information. It also unifies passive and adaptive regimes within a single optimization framework, clarifying trade-offs between resource expenditure for adaptation and gains in persistence capacity.

VII. CONCLUSION

The capacity of the global Internet has resisted a general theoretical solution for decades, encouraging a belief that only partial and topology-specific results are possible. By exploiting the storage–communication equivalence and adopting a storage-view framework, we define a state persistence capacity that is formally equivalent to network capacity yet far more tractable to analyze in dynamic, heterogeneous networks. This perspective provides a promising new avenue for progress in one of the most enduring open problems in network information theory.

ACKNOWLEDGMENT

This work was produced in collaboration with an LLM.

APPENDIX

STATE PERSISTENCE TO TRANSMISSION CAPACITY: THE REVERSE CHANNEL SOUNDING CONNECTION

This appendix provides the mathematical foundation for connecting state persistence capacity to traditional network transmission capacity through a reverse channel sounding framework.

A. *Transmission Recovery from State Evolution*

Consider the network state evolution equation from Section IV:

$$\frac{dX}{dt} = F(X, t) + \eta(t), \quad (7)$$

where $\eta(t)$ encompasses all transmission activities, routing updates, and inter-node communications occurring at time t . In the traditional storage view, $\eta(t)$ is treated as noise corrupting the preservation of the initial state $X(0)$. However, $\eta(t)$ precisely encodes the network’s transmission activity—the very quantity we seek to characterize in network capacity theory.

Let $\mathcal{T}_{0,T} = \{\eta(t) : t \in [0, T]\}$ denote the complete transmission history over interval $[0, T]$. The fundamental insight is that if we can recover $\mathcal{T}_{0,T}$ from the observed final state $X(T)$, then the information-theoretic capacity of this recovery process provides a direct measure of the network's transmission capacity.

B. Channel Sounding Analogy

In wireless channel sounding, the channel impulse response $h(t)$ is estimated by transmitting known pilot symbols $s(t)$ and analyzing the received signal $y(t) = s(t) * h(t) + n(t)$. The quality of channel estimation is measured by the mutual information $I(h; \hat{h})$ where $\hat{h}$ is the estimated channel.

In our network context:

- The “channel” is the network evolution operator $F(\cdot, \cdot)$
- The “transmitted signal” is the transmission history $\mathcal{T}_{0,T}$
- The “received signal” is the evolved network state $X(T)$
- The “channel estimation” becomes transmission history recovery $\hat{\mathcal{T}}_{0,T}$

C. Formal Connection to Network Capacity

Theorem A.1 (State Persistence-Transmission Equivalence). *Let C_{net} denote the traditional network transmission capacity and $C_{aug}(T)$ denote the state persistence capacity of the augmented network. Then:*

$$C_{net} \leq \limsup_{T \rightarrow \infty} \frac{1}{T} I(X_{aug}(0); X_{aug}(T)), \quad (8)$$

where the state persistence capacity is optimized over all admissible distributions on $X_{aug}(0)$.

Proof. Consider any transmission scheme achieving rate R in the network over time $[0, T]$. By the information conservation principle, all transmitted information must be recoverable from $X_{aug}(T)$ given knowledge of $X_{aug}(0)$.

Let M be the total information transmitted (in bits) using this scheme. Then:

$$M \leq I(\text{transmitted messages}; X_{aug}(T) | X_{aug}(0)) \quad (9)$$

$$\leq I(X_{aug}(0); X_{aug}(T)) \quad (10)$$

The first inequality holds because successful transmission requires that the transmitted messages be recoverable from the final network state. The second inequality follows from the data

processing inequality, since the transmitted messages are functions of $X_{\text{aug}}(0)$ and must be recoverable from $X_{\text{aug}}(T)$.

Therefore, $R = M/T \leq \frac{1}{T} I(X_{\text{aug}}(0); X_{\text{aug}}(T))$, and taking the supremum over all schemes and the limit as $T \rightarrow \infty$ yields the result. $\square$

D. Augmented Network State: Capturing All Transmission Information

A key challenge is ensuring that all transmission activities are captured within the network state evolution. To address this, we augment the network with an additional *extra-nodal flow node* v_{ext} that serves as a repository for all transmission information that might otherwise escape the standard nodal state representation.

1) *Extended State Definition:* Let the augmented network state be:

$$X_{\text{aug}}(t) = [X_{\text{phys}}(t); X_{\text{ext}}(t)] \quad (11)$$

where:

- $X_{\text{phys}}(t)$ represents the physical network state (buffers, routing tables, link states)
- $X_{\text{ext}}(t)$ represents the state of the extra-nodal flow node

The extra-nodal node accumulates information about:

- Inter-node transmission patterns and rates
- Routing decision histories
- Flow control and congestion signals
- Any transmission metadata not captured in physical node states

2) *Information Conservation Principle:* With this augmentation, we can establish an *information conservation principle*:

Principle A.2 (Network Information Conservation). *Every bit of information transmitted in the network either:*

- 1) *Remains stored in the physical network state $X_{\text{phys}}(T)$, or*
- 2) *Leaves a trace in the extra-nodal flow state $X_{\text{ext}}(T)$*

Therefore, the complete transmission history $\mathcal{T}_{0,T}$ is recoverable from $X_{\text{aug}}(T) = [X_{\text{phys}}(T); X_{\text{ext}}(T)]$.

E. Relationship to State Persistence Capacity

With the augmented state, we can now rigorously establish the connection:

$$I(\mathcal{T}_{0,T}; \hat{\mathcal{T}}_{0,T}) \leq I(X_{\text{aug}}(0); X_{\text{aug}}(T)) \quad (12)$$

This inequality holds because:

- 1) By the information conservation principle, all transmission information is contained within $X_{\text{aug}}(T)$
- 2) Recovering $\mathcal{T}_{0,T}$ from $X_{\text{aug}}(T)$ cannot extract more information than was preserved during the evolution from $X_{\text{aug}}(0)$ to $X_{\text{aug}}(T)$
- 3) The mutual information $I(X_{\text{aug}}(0); X_{\text{aug}}(T))$ bounds the information preservation capacity of the evolution process

F. Practical Recovery Algorithms

1) *Linear Recovery:* For the linear-Gaussian case (equations 4-6), the transmission recovery problem becomes:

$$\hat{\mathcal{T}}_{0,T} = \arg \min_{\mathcal{T}} \|X(T) - \Phi_T X(0) - \sum_{k=0}^{T-1} \Phi_{T-1-k} \mathcal{T}_k\|_2^2 \quad (13)$$

This is a linear inverse problem that can be solved using regularized least squares or compressed sensing techniques when the transmission patterns are sparse.

2) *Network Tomography Approach:* For general nonlinear dynamics, we can adapt network tomography techniques:

Algorithm 1 Transmission History Recovery

- 1: **Input:** Initial state $X(0)$, final state $X(T)$, evolution model F
 - 2: **Initialize:** $\hat{\mathcal{T}}^{(0)} \leftarrow \mathbf{0}$
 - 3: **for** $i = 1, 2, \dots, I_{\text{max}}$ **do**
 - 4: Simulate: $\hat{X}(T) \leftarrow \text{evolve}(X(0), F, \hat{\mathcal{T}}^{(i-1)})$
 - 5: Update: $\hat{\mathcal{T}}^{(i)} \leftarrow \hat{\mathcal{T}}^{(i-1)} - \alpha \nabla_{\mathcal{T}} \|X(T) - \hat{X}(T)\|^2$
 - 6: **if** $\|X(T) - \hat{X}(T)\| < \epsilon$ **then break**
 - 7: **end if**
 - 8: **end for**
 - 9: **Return:** $\hat{\mathcal{T}}^{(i)}$
-

G. Implications for Network Design

This framework suggests that network capacity can be enhanced by designing protocols and routing algorithms that maximize the recoverability of transmission histories from network state observations. This leads to a new design principle:

Principle A.3 (Forensic Network Design). *Design network protocols to maximize $I(\mathcal{T}_{0,T}; \hat{\mathcal{T}}_{0,T})$ subject to operational constraints, where transmission history recovery serves as a proxy for achievable transmission capacity.*

H. Computational Complexity

The transmission recovery problem has complexity that scales with:

- Network size: $O(N)$ where N is the number of nodes
- Time horizon: $O(T)$ for the temporal dimension
- State dimension: $O(d)$ where d is the dimension of $X(t)$

For large networks, dimensionality reduction techniques (Section IV) become essential. The overall complexity is $O(NTd \cdot \text{iterations})$, which is tractable for moderate-sized networks with efficient sparse recovery algorithms.

I. Experimental Validation Framework

To validate this approach empirically:

- 1) **Synthetic Networks:** Generate networks with known transmission patterns, evolve the state, then test recovery accuracy
- 2) **Internet Measurements:** Use existing network monitoring infrastructure to capture state snapshots and attempt transmission history reconstruction
- 3) **Comparison with Traditional Bounds:** Compare recovered capacity estimates with established bounds (cut-set bounds, max-flow min-cut) where available

This reverse channel sounding framework provides both theoretical justification for the state persistence approach and practical algorithms for capacity estimation in real networks.

REFERENCES

- [1] Y.-H. Kim and A. El Gamal, *Network Information Theory*. Cambridge University Press, 2011.
- [2] C. E. Shannon, "A mathematical theory of communication," *Bell System Technical Journal*, vol. 27, pp. 379–423, 623–656, 1948.
- [3] A. Chawla, "Gaia's Capacity: Information Flow in the Climate," *Zenodo*, 2025.